Benchmarking Framework for Large-Scale Modular Quantum Systems

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Abstract—Credible quantum computing programs require benchmark frameworks that distinguish measured results from architectural projections and aspirational targets. This paper defines the QONTOS benchmarking framework: a layered protocol suite that spans simulator-backed orchestration, digital-twin modeling, noisy baselines, and eventual fault-tolerant modular hardware. We specify exact benchmark definitions grounded in the peer-reviewed literature—including randomized benchmarking, Quantum Volume, CLOPS, and application-oriented suites—together with measurement conditions, confidence-interval reporting requirements, baseline comparisons, and pass/fail gating thresholds. The framework classifies every reported metric by its evidence level using mandatory claim labels and establishes a modular-system benchmarking methodology for the QONTOS hybrid superconducting-photonic architecture, covering both intra-module (superconducting) performance and cross-module (photonic) metrics such as inter-module gate fidelity and Bell pair generation rates. All protocols reference the open-source qontos-benchmarks repository as the canonical implementation. This paper serves as the credibility backbone of the QONTOS paper series: no claim in any companion paper is valid unless it maps to a benchmark, protocol, or validation gate defined here.

Claim label: METHODOLOGY—benchmark design, protocol specification, and release gating framework. No performance claims are made in this paper; all numeric values appearing in target tables carry explicit QONTOS-TARGET or STRETCH-TARGET labels.

I. Introduction: Benchmarking as the Credibility Engine

A. Motivation and Scope

The quantum computing field suffers from a credibility gap. Headline metrics are frequently reported without specifying measurement conditions, noise models, or the boundary between measured values and architectural projections. Proctor et al. (2022) demonstrated that even well-intentioned benchmark reporting can mislead when measurement protocols are underspecified [4]. Lubinski et al. (2023) further showed that application-oriented benchmarks require careful normalization to enable meaningful cross-platform comparison [5].

[CLAIM-METHODOLOGY] The QONTOS benchmarking framework addresses this gap by requiring every reported metric to carry: (a) a claim label specifying its evidence basis, (b) a complete measurement-condition record, (c) at least one baseline comparison, and (d) a confidence interval or explicit uncertainty statement.

This paper serves a structural role in the QONTOS paper series. Papers 01 through 08 and Paper 10 contain architectural claims, algorithm projections, and roadmap targets. None of those claims are valid in isolation. Each must trace to a benchmark definition, measurement protocol, or validation gate specified here. If a claim cannot be mapped to this framework, it must be labeled UNVALIDATED until a corresponding benchmark is defined.

B. Claim Label Taxonomy

Every metric, result, or projection reported in any QONTOS paper must carry exactly one of the following labels:

TABLE I
Claim label taxonomy.

Claim Label	Definition
MEASURED-SIMULATOR	Result obtained on the QONTOS simulator-backed pipeline
MEASURED-HARDWARE	Result obtained on physical quantum hardware
DIGITAL-TWIN-PROJECTION	Result from a calibrated noise model and modular runtime simulation
NOISY-BASELINE	Result from a standardized depolarizing or realistic noise channel
QONTOS-TARGET	Planned performance level under stated assumptions
STRETCH-TARGET	Aspirational performance level requiring multiple breakthroughs
LITERATURE-VALUE	Value taken from a published, peer-reviewed source
UNVALIDATED	Claim that does not yet map to a benchmark protocol

C. Benchmark Definitions

This section defines the core benchmarks used throughout the QONTOS program, grounded in their original peer-reviewed specifications.

Randomized Benchmarking (RB). Introduced by Knill et al. (2008) [3], randomized benchmarking estimates average gate fidelity by measuring the decay of sequence fidelity as a function of the number of random Clifford gates applied to a qubit register. The protocol is efficient because the number of required sequences scales polynomially, and the result is largely independent of state-preparation and measurement (SPAM) errors. QONTOS

adopts standard RB as the Layer-0 single-qubit and two-qubit gate characterization protocol.

Interleaved Randomized Benchmarking (IRB). Magesan et al. (2012) [2] extended RB to isolate the error rate of a specific gate by interleaving it between random Clifford operations. The ratio of the interleaved and reference decay rates yields the error per gate for the target operation. QONTOS uses IRB to characterize individual gate types—particularly cross-module entangling gates whose fidelity is critical to the modular architecture (see Paper 01).

Quantum Volume (QV). Defined by Cross et al. (2019) [1], Quantum Volume measures the largest square circuit (depth = width = d) that a quantum computer can execute with a heavy-output-generation probability exceeding 2/3. The protocol involves sampling random SU(4) circuits, computing the heavy-output set classically, and verifying that the quantum device's heavy-output fraction passes a one-sided binomial test at the 97.7% confidence level (two sigma). IBM (2024) has maintained and updated the operational definition [6]. QONTOS reports QV at each benchmark layer with the full Cross et al. protocol, including the confidence-level requirement.

Circuit Layer Operations Per Second (CLOPS). Introduced by Wack et al. (2021) [7], CLOPS measures the throughput of a quantum computing system by counting the number of QV-equivalent circuit layers that can be executed per second, including compilation, execution, and result retrieval. CLOPS captures the full-stack latency that determines practical utility. QONTOS reports CLOPS with explicit decomposition into compilation time, queue time, execution time, and classical post-processing time.

Application-Oriented Benchmarks. Lubinski et al. (2023) [5] defined a suite of application-oriented benchmarks spanning chemistry (VQE), optimization (QAOA, MaxCut), machine learning (quantum kernel methods), and simulation (Hamiltonian dynamics). Each benchmark specifies a problem family, a scaling parameter, and a figure of merit. QONTOS adopts this framework for its application benchmark ladder (Section 5).

SupermarQ. Tomesh et al. (2022) [8] introduced a scalable benchmark framework that decomposes quantum program performance into communication, computation, critical depth, entanglement, and parallelism features. QONTOS uses SupermarQ feature vectors to characterize workload difficulty independently of hardware.

D. Measurement Conditions

[CLAIM-METHODOLOGY] Every reported benchmark result must include the following metadata record. Results lacking any required field are classified UNVALIDATED.

TABLE II
Required measurement condition fields.

Field	Req.	Description
claim_label	Yes	One of the labels from Section 1.2
benchmark_id	Yes	Canonical identifier from repo
benchmark_layer	Yes	L0, L1, L2, or L3
execution_environment	Yes	Simulator, digital twin, or HW
workload_specification	Yes	Circuit family, width, depth
noise_model	Yes	Exact noise channel or "none"
hardware_assumptions	Yes	Phys. qubit count, connectivity
modular_assumptions	Cond.	Module count, link fidelity
reconstruction_method	Cond.	If post-processing affects metric
confidence_interval	Yes	95% CI or uncertainty statement
software_version	Yes	Git commit hash
timestamp	Yes	ISO 8601 execution timestamp
reproducibility_hash	Yes	SHA-256 of run configuration

II. Benchmark Layers and Modular-System Benchmarking

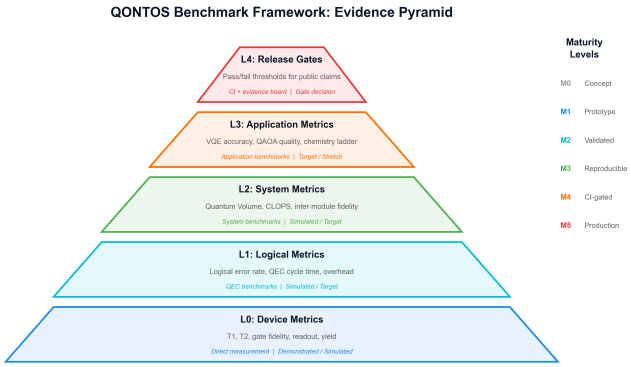

Fig. 1. Benchmark Evidence Pyramid.

A. Layer Definitions

QONTOS organizes benchmarks into four layers reflecting the maturity of the execution environment. Because QONTOS builds a hybrid superconducting-photonic modular architecture—combining high-coherence tantalum-on-silicon transmon qubits (Paper 02) with photonic interconnects for inter-module entanglement distribution (Paper 04)—the benchmark framework must capture performance along both the superconducting device axis and the photonic networking axis.

TABLE III
Benchmark layer definitions.

Layer	Environment	Evidence	Use
L0	Simulator-backed pipeline	MEAS.-SIM.	Platform validation
L1	Digital-twin runtime	DT-PROJ.	Architecture projections
L2	Prototype HW	MEAS.-HW	Device characterization
L3	Large-scale FT system	TARGET	Roadmap milestones

Layer promotion rule. A metric may only advance from a lower layer to a higher layer when the evidence requirements of the higher layer are fully satisfied. A digital-twin projection (L1) does not become a hardware result (L2) merely by asserting that the noise model is realistic. Hardware measurement on a calibrated device with raw data is required.

B. Benchmark Protocol Specification

Each benchmark in the qontos-benchmarks repository is defined by a protocol file. For example, the Quantum Volume protocol (QV-STANDARD-001) specifies:

1) Select target width d.
2) Generate $N \geq 100$ random SU(4) circuits of depth d on d qubits.
3) For each circuit, compute the ideal heavy-output set classically.
4) Execute each circuit on the target backend with $S \geq 100$ shots.
5) Compute the heavy-output fraction h_i for each circuit.
6) Compute $\bar{h} = (1/N) \sum h_i$.
7) Apply one-sided binomial test: reject H_0 ($\bar{h} \leq 2/3$) at $\alpha = 0.023$.
8) Report QV $= 2^d$ if test passes; otherwise report QV $< 2^d$.

Pass threshold: $\bar{h} > 2/3$ with 97.7% confidence. Confidence interval: 95% bootstrap CI over N circuits.

[CLAIM-METHODOLOGY] This protocol format is mandatory for all benchmarks. The qontos-benchmarks repository contains the reference implementation of every protocol defined in this paper.

C. Modular-System Benchmarking

Modular quantum architectures introduce benchmark challenges absent from monolithic systems. In the QONTOS hybrid superconducting-photonic platform, cross-module metrics are as critical as intra-module metrics. Martiel et al. (2021) [9] identified coprocessor-level benchmarking as a distinct problem requiring metrics that capture inter-module communication overhead.

[CLAIM-METHODOLOGY] QONTOS defines the following modular-system benchmark extensions:

Inter-module gate fidelity. Measured via IRB (Magesan et al. 2012 [2]) applied specifically to entangling gates that span module boundaries.

Modular Quantum Volume (MQV). Extension of the QV protocol where the d-qubit register is distributed across k modules. MQV is reported as a pair $(\mathrm{QV}_{\mathrm{value}}, k_{\mathrm{modules}})$.

Distributed CLOPS (D-CLOPS). Extension of CLOPS where circuit layers include inter-module communication steps. D-CLOPS is always reported alongside monolithic CLOPS on the same workload.

Cross-module entanglement rate. The number of Bell pairs generated per second across a module boundary, measured with full tomographic verification.

Module-aware circuit partitioning overhead. The ratio of total two-qubit gates after partitioning to the minimum two-qubit gate count on an all-to-all connected ideal machine.

III. Baseline Comparisons and Gating Thresholds

A. Three Mandatory Baselines

[CLAIM-METHODOLOGY] Every benchmark result at any layer must be reported alongside at least one of the following baselines. L2 and L3 results require all three.

Ideal simulator baseline. The benchmark is executed on a noiseless statevector or tensor-network simulator with the same circuit, qubit count, and measurement configuration.

Noisy baseline. The benchmark is executed on a standardized noise model: symmetric depolarizing noise at error rate $p = 10^{-3}$ per single-qubit gate and $p = 10^{-2}$ per two-qubit gate, with $T_1 = 100$ µs and $T_2 = 100$ µs relaxation.

Digital-twin baseline. The benchmark is executed on a calibrated noise model derived from the most recent hardware characterization data.

B. Baseline Comparison Protocol

Results must be presented in the following tabular format:

TABLE IV
Baseline comparison format.

Metric	Ideal	Noisy	DT	Meas.	Label
QV (d=5)	32	32	32	—	MEAS.-SIM.
QV (d=10)	1024	256	512	—	DT-PROJ.
CLOPS	—	—	—	—	TARGET

[CLAIM-METHODOLOGY] The "Measured Value" column remains empty ("—") until a hardware measurement is obtained. Under no circumstances may a digital-twin projection be placed in the "Measured Value" column.

C. Pass/Fail Thresholds

Each benchmark layer has explicit gating thresholds.

Layer L0 gates:

TABLE V
Layer L0 gating thresholds.

Gate	Requirement	Pass Criterion
G0-1	Protocol implemented	Protocol file exists, tests pass
G0-2	Ideal baseline correct	Deviation $< 3\sigma$ from theory
G0-3	Noisy baseline expected	Within analytic bound $\pm 10\%$
G0-4	Fully reproducible	Re-execution within CI

Layer L1 gates:

TABLE VI
Layer L1 gating thresholds.

Gate	Requirement	Pass Criterion
G1-1	All L0 gates passed	Documented
G1-2	Noise model sourced	Provenance documented
G1-3	Modular assumptions stated	Module count, topology recorded
G1-4	DT bounded by baselines	ideal $\geq$ DT $\geq$ noisy

Layer L2 gates:

TABLE VII
Layer L2 gating thresholds.

Gate	Requirement	Pass Criterion
G2-1	All L1 gates passed	Documented
G2-2	HW calibration available	Within 24 hours
G2-3	Raw data archived	Data hash recorded
G2-4	Consistent with DT	Within 3σ or explained

Layer L3 gates:

TABLE VIII
Layer L3 gating thresholds.

Gate	Requirement	Pass Criterion
G3-1	All L2 gates passed per module	Documented per module
G3-2	Modular benchmarks reported	MQV, D-CLOPS present
G3-3	Application benchmark done	Above current validated level
G3-4	FT overhead measured	Decoder latency included

IV. Current QONTOS Benchmark Surface

A. Current Benchmark Harness

[CLAIM-MEASURED-SIMULATOR] The QONTOS benchmark harness is implemented in the qontos-benchmarks repository and currently supports the following capabilities:

TABLE IX
Current benchmark harness status.

Capability	Status
Randomized benchmarking (RB) [3]	Implemented
Interleaved RB [2]	Implemented
Quantum Volume [1]	Implemented
CLOPS [7]	Implemented
Application benchmarks (H_2, LiH) [5]	Implemented
Small QAOA benchmarks	Implemented
Modular QV (MQV)	In development
Distributed CLOPS	In development
SupermarQ feature extraction [8]	Planned

Execution environments currently supported:

- Statevector simulator: Ideal noiseless execution for baseline generation. Up to 30 qubits on standard hardware, up to 40 qubits on high-memory nodes.
- Noisy simulator: Depolarizing and relaxation noise models with configurable parameters.
- Digital-twin runtime: Modular architecture simulator with configurable module count, topology, inter-module gate fidelity, and classical communication latency.

B. Benchmark Report Format

Every benchmark execution produces a structured report containing the report ID, benchmark ID, execution timestamp, software version, environment, configuration hash, baselines (ideal, noisy, digital-twin), result with confidence interval, claim label, and gates passed/failed.

V. Application Benchmark Ladder

[CLAIM-METHODOLOGY] QONTOS benchmarks application-level progress using graduated workload ladders rather than a single flagship endpoint.

A. Chemistry Ladder

TABLE X
Chemistry application benchmark ladder.

Rung	Workload	Qubits	Threshold	Claim
C1	H_2 ground state	2–4	<1.6 mHa	MEAS.-SIM.
C2	LiH dissociation	6–12	<1.6 mHa	MEAS.-SIM.
C3	Small catalysts	20–40	<4 mHa	DT-PROJ.
C4	Medium catalysts	50–100	<10 mHa	TARGET
C5	FeMoco-class	100–200	<1.6 mHa	STRETCH

[CLAIM-METHODOLOGY] FeMoco (rung C5) remains the flagship stretch benchmark, but no QONTOS communication may reference FeMoco without also reporting current validated performance on rungs C1 and C2.

B. Optimization Ladder

TABLE XI
Optimization application benchmark ladder.

Rung	Workload	Qubits	Threshold	Claim
O1	MaxCut ($n \leq 16$)	8–16	Ratio > 0.8	MEAS.-SIM.
O2	Portfolio opt.	15–30	Within 5%	DT-PROJ.
O3	Vehicle routing	50–100	Competitive TTS	TARGET
O4	Large-scale	1000+	Measurable adv.	STRETCH

C. Simulation Ladder

TABLE XII
Simulation application benchmark ladder.

Rung	Workload	Qubits	Threshold	Claim
S1	Ising dynamics	10–20	$F > 0.95$	MEAS.-SIM.
S2	Heisenberg	25–50	Error < 5%	DT-PROJ.
S3	Lattice gauge	100+	Controlled err.	STRETCH

D. Flagship Benchmark Rule

[CLAIM-METHODOLOGY] No flagship benchmark (C5, O4, S3) may appear in any QONTOS communication without: (1) current validated rung on the same ladder, (2) number of rungs between current capability and the flagship, (3) explicit list of requirements for each intermediate rung, and (4) estimated timeline with uncertainty range.

VI. System-Level Metric Definitions

A. Gate-Level Metrics

TABLE XIII
Gate-level metric definitions.

Metric	Protocol	Reference
1Q gate error	Standard RB	Knill et al. [3]
2Q gate error	IRB	Magesan et al. [2]
Cross-module gate error	IRB (modular)	Magesan et al. [2]
SPAM error	Dedicated protocol	Knill et al. [3]

B. System-Level Metrics

TABLE XIV
System-level metric definitions.

Metric	Protocol	Reference
Quantum Volume	Heavy-output test	Cross et al. [1]
Modular QV	Distributed test	This paper
CLOPS	QV-layer throughput	Wack et al. [7]
D-CLOPS	Modular throughput	This paper

C. Logical-Qubit Metrics

[CLAIM-METHODOLOGY] A logical qubit is counted as usable only when all of the following are specified: (1) error-correcting code family, (2) code distance and physical-to-logical ratio, (3) decoder type and latency, (4) logical error rate per cycle with cycle duration, and (5) correction cycle budget for the target computation.

TABLE XV
Logical-qubit metric targets.

Metric	Base	Aggressive	Stretch
Phys.-to-logical ratio	1000:1	300:1	100:1
Logical error rate	10^{-4}/cycle	10^{-6}/cycle	10^{-8}/cycle
Decoder latency	< 1 µs	< 0.5 µs	< 0.1 µs
Logical qubit count	1–20	100–1,000	10,000+

D. Scenario Targets (System Level)

[CLAIM-QONTOS-TARGET / STRETCH-TARGET]

TABLE XVI
System-level scenario targets.

Metric	L0	L1	L2	L3
QV	Sim.	2^{10}–2^{15}	2^{15}–2^{20}	2^{20}+
CLOPS	Sim.	10^3–10^5	10^5–10^8	10^8–10^{12}
Logical qubits	N/A	1–20	100–1,000	10,000+
Bell rate	N/A	10^3/s	10^6/s	10^9/s
Circuit depth	10^3	10^4	10^6	10^{12}

VII. Confidence Interval and Statistical Reporting

A. Mandatory Statistical Requirements

[CLAIM-METHODOLOGY] All benchmark results must report uncertainty using the following conventions:

1) Quantum Volume: One-sided binomial test at $\alpha = 0.023$ (97.7% confidence), per Cross et al. (2019) [1]. Additionally report the 95% bootstrap CI for the mean heavy-output fraction.
2) CLOPS: Report mean and standard deviation over at least 10 independent executions. Report 95% CI using the t-distribution.
3) Gate fidelity (RB/IRB): Report the fit uncertainty from the exponential decay model. Use at least 30 random sequences per sequence length and at least 6 sequence lengths. Report 95% CI on the error-per-gate parameter.
4) Application benchmarks: Report the mean and 95% CI of the figure of merit over at least 10 independent runs with different random seeds or shot allocations.

B. Significance Testing for Improvement Claims

[CLAIM-METHODOLOGY] Any claim that a new QONTOS version, algorithm, or hardware configuration improves benchmark performance must be supported by a two-sided hypothesis test at the $\alpha = 0.05$ significance level. The test must account for multiple comparisons if more than one metric is simultaneously evaluated (Bonferroni correction or equivalent).

VIII. Release and Communication Gating

A. Benchmark Maturity Levels

TABLE XVII
Benchmark maturity levels.

Stage	Name	Requirements
M0	Protocol draft	Protocol file written; not implemented
M1	Harness implemented	Code in repo; unit tests pass
M2	Baselines validated	All baselines produce expected results
M3	Report generated	All metadata fields populated
M4	Peer reviewed	Reviewed by non-involved team member
M5	Publication ready	All gates passed; archived with hash

B. Safe Communication Language

The following language is pre-approved for external communication:

- "QONTOS has measured [metric] on its simulator-backed pipeline"—requires M5, MEASURED-SIMULATOR.
- "QONTOS projects [metric] under [assumptions] using its digital twin"—requires M5, DIGITAL-TWIN-PROJECTION.
- "QONTOS targets [metric] for its [timeline] modular architecture"—requires M3, QONTOS-TARGET.
- "QONTOS has defined a stretch target of [metric] requiring [breakthroughs]"—requires M3, STRETCH-TARGET.

C. Prohibited Communication Language

The following formulations are prohibited unless the corresponding MEASURED-HARDWARE result at M5 exists: "QONTOS achieves Quantum Volume X," "QONTOS demonstrates CLOPS of Y," "QONTOS has Z logical qubits operational," "QONTOS demonstrates quantum advantage," "QONTOS outperforms [competitor] on [benchmark]."

IX. Connection to the QONTOS Paper Series

This benchmarking framework provides the validation backbone for every companion paper:

TABLE XVIII
Paper series benchmark connections.

Paper	Key Claims Validated Here
01: Architecture	Module count targets, inter-module fidelity
02: Ta–Si Qubits	Gate fidelity, T_1/T_2 protocols
03: Error Correction	Logical error rate, overhead ratios
04: Photonic Links	Cross-module Bell rate, gate fidelity
05: AI Decoding	Decoder latency, logical error rate
06: Software Stack	CLOPS, partitioning overhead
07: Cryogentic Infra.	HW calibration for digital-twin baselines
08: Algorithms	Application benchmark ladder
10: Roadmap 2030	Scenario targets, milestone gating

[CLAIM-METHODOLOGY] If a claim in any companion paper cannot be traced to a benchmark definition, measurement condition, or validation gate in this paper, that claim must carry the UNVALIDATED label.

X. Conclusion

Benchmarking is not a post-hoc validation exercise—it is the structural foundation on which every QONTOS claim stands or falls. This paper has established:

1) A claim-label taxonomy (Section 1.2) that forces every metric to declare its evidence basis.
2) Rigorous benchmark definitions (Section 1.3) grounded in peer-reviewed protocols from Knill et al. [3], Magesan et al. [2], Cross et al. [1], Wack et al. [7], Lubinski et al. [5], and Tomesh et al. [8].
3) Mandatory measurement conditions (Section 1.4) ensuring reproducibility.
4) Modular-system extensions (Section 2.3) including MQV and D-CLOPS.
5) Three mandatory baselines (Section 3.1) against which every result must be compared.
6) Explicit pass/fail gates (Section 3.3) preventing premature promotion.
7) Application benchmark ladders (Section 5) ensuring visibility of the gap to flagship targets.
8) Statistical reporting requirements (Section 7) including confidence intervals and significance testing.
9) Communication gating (Section 8) preventing unsafe language.

The most important contribution is the structural discipline: every claim maps to a protocol, every protocol maps to an implementation, every implementation produces baselines, and every result carries a confidence interval and a claim label.

[CLAIM-METHODOLOGY] All benchmark protocols, harness code, baseline configurations, and report templates are maintained in the qontos-benchmarks repository. This paper and the repository evolve together.

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